

Cisco Certified Network Associate (CCNA)



The **Cisco Certified Network Associate (CCNA)** is an entry-to-associate level networking certification from **Cisco**.

It validates skills required to install, configure, operate, troubleshoot, and secure networks.

It is one of the most recognized certifications for Network Engineers, System Administrators, and Network Support Engineers.

1. Basic Overview

Item	Details
Certification Name	CCNA
Exam Code	200-301
Vendor	Cisco
Level	Associate

Duration

120 minutes

Questions

~100–120

Cost

~\$300 USD

Validity

3 years

Format

MCQ, drag-drop, simulations

2. Skills You Learn in CCNA

CCNA focuses on **practical networking skills**.

Main capabilities:

- Network fundamentals
- Routing and switching
- IP addressing
- VLAN configuration
- Network security basics
- Automation and programmability
- Troubleshooting networks

3. CCNA Exam Syllabus

Network Fundamentals (20%)

Topics include:

- OSI Model
- TCP/IP Model
- Network components
- IPv4 and IPv6
- Subnetting
- ARP
- MAC address
- Ethernet
- Cabling types
- Wireless basics

2 Network Access (20%)

Focuses on switching technologies.

Topics:

- VLAN
- Trunking (802.1Q)
- Inter-VLAN routing
- Spanning Tree Protocol (STP)
- EtherChannel
- Wireless LAN concepts
- Switch configuration

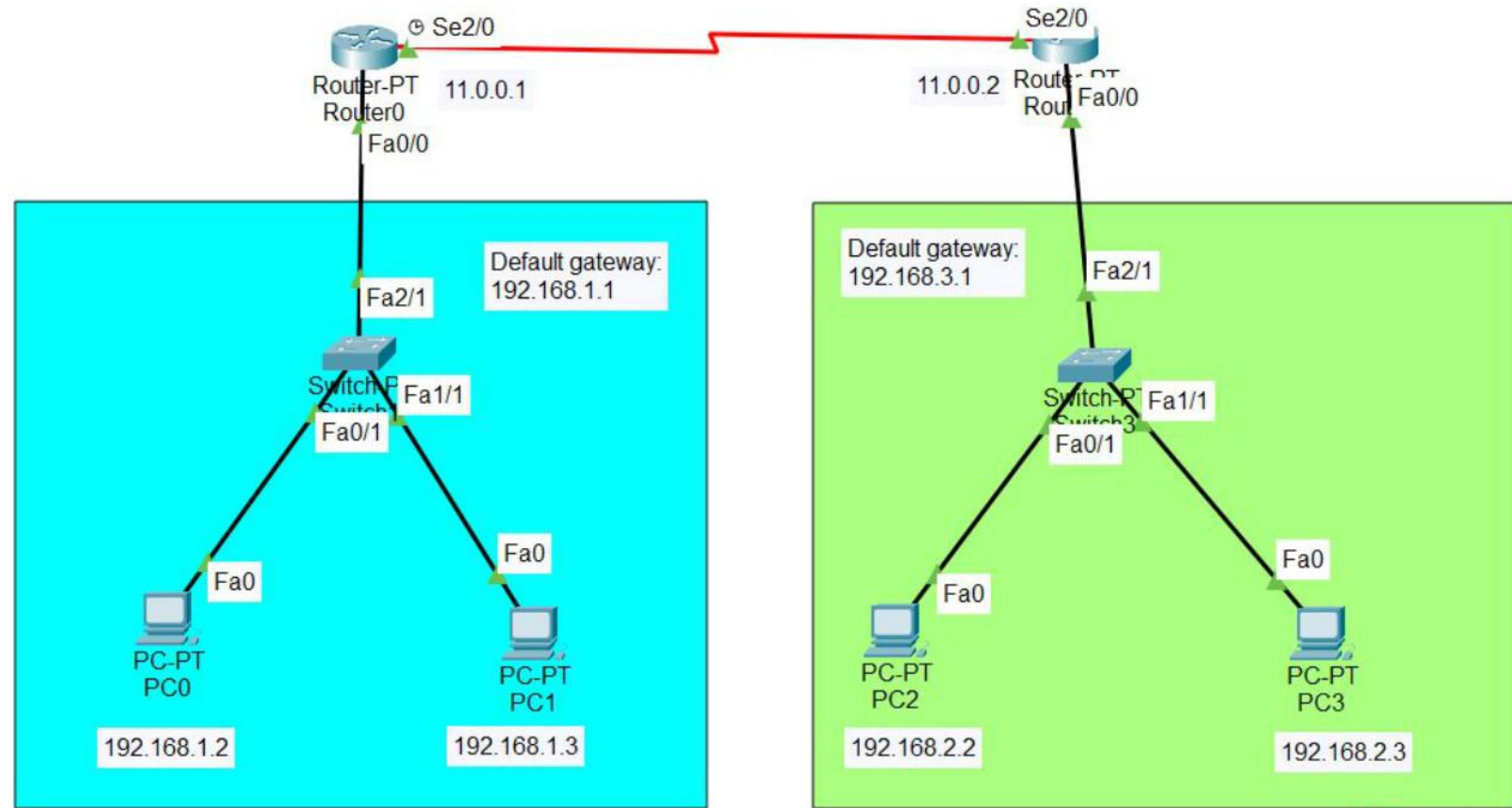


3 IP Connectivity (25%)

Core routing topics.

Topics include:

- Static routing
- Default route
- Dynamic routing
- OSPF (single area)
- Routing tables
- Packet forwarding

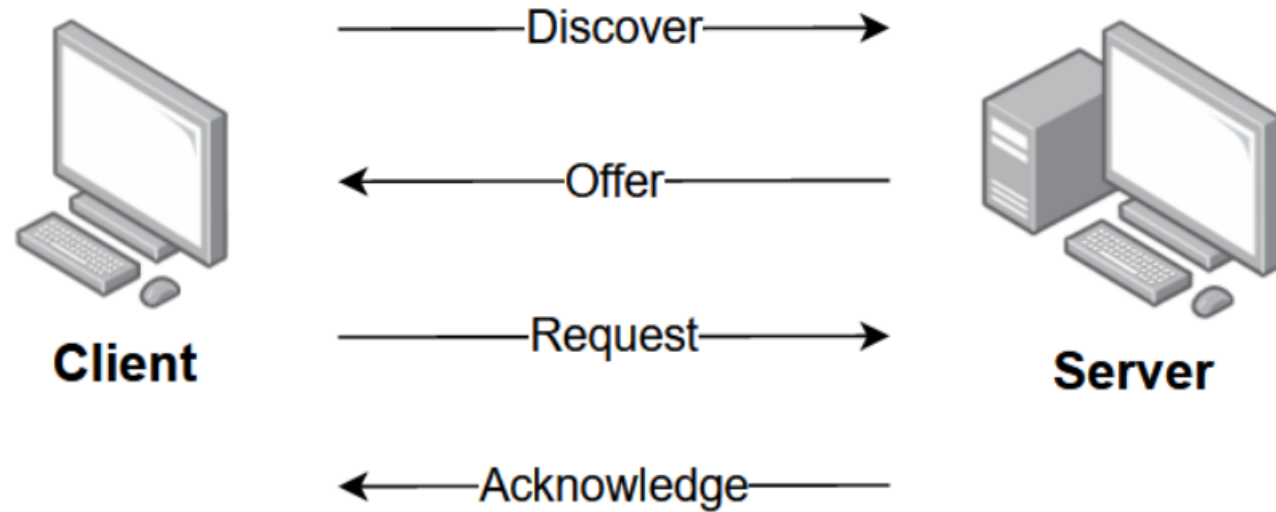


4 IP Services (10%)

Network services used in real environments.

Topics:

- DHCP
- DNS
- NTP
- NAT / PAT
- SNMP



5 Security Fundamentals (15%)

Basic security practices.

Topics:

- Access Control Lists (ACL)
- Port Security
- SSH
- Device hardening
- Network attacks
- VPN basics

Automation and Programmability (10%)

Modern networking concepts.

Topics:

- Software-Defined Networking (SDN)
- APIs
- JSON
- REST
- Configuration management tools
- Controller-based networking

4. Labs Tools for Practice

Cisco Packet Tracer

Best for beginners.

Features:

- Simulated routers and switches
- Packet simulation
- Easy topology building

5. Jobs After CCNA

Common roles:

- Network Engineer
- Network Administrator
- NOC Engineer
- System Administrator
- IT Support Engineer

6. Career Path After CCNA

Networking certification path:

CCNA → CCNP → CCIE

Advanced options:

- Cloud Networking
- Network Security
- Automation (Python + NetDevOps)